Brennan Duff
Assignment 5

1.

bookid	title	author	category	price
1	The Hobbit	J.R.R. Tolkien	Fantasy	15.99
2	1984	George Orwell	Dystopian	12.5
3	To Kill a Mockingbird	Harper Lee	Fiction	14.25
4	Dune	Frank Herbert	Sci-Fi	18
5	Foundation	Isaac Asimov	Sci-Fi	16.5
6	Pride and Prejudice	Jane Austen	Romance	10.99
7	The Great Gatsby	F. Scott Fitzgerald	Classic	11.5
8	Moby-Dick	Herman Melville	Adventure	13.99
9	War and Peace	Leo Tolstoy	Historical	19.99
10	The Odyssey	Homer	Classic	12.99
11	Crime and Punishm...	Fyodor Dostoevsky	Classic	14.5
12	The Catcher in the ...	J.D. Salinger	Fiction	10.5
13	Brave New World	Aldous Huxley	Dystopian	13.5
14	The Lord of the Rings	J.R.R. Tolkien	Fantasy	25.99
15	Fahrenheit 451	Ray Bradbury	Dystopian	11.99
16	Jane Eyre	Charlotte Bronte	Romance	12
17	The Chronicles of ...	C.S. Lewis	Fantasy	21.5
18	Animal Farm	George Orwell	Dystopian	9.99
19	The Alchemist	Paulo Coelho	Adventure	14
20	The Little Prince	Antoine de Saint-...	Fable	9.5
21	Dracula	Bram Stoker	Horror	11.25
22	Frankenstein	Mary Shelley	Horror	10.75
23	The Picture of Dori...	Oscar Wilde	Classic	11
24	Les MisÃ©rables	Victor Hugo	Historical	18.5
25	The Count of Mont...	Alexandre Dumas	Adventure	16
26	The Brothers Kara...	Fyodor Dostoevsky	Classic	17.5
27	Anna Karenina	Leo Tolstoy	Historical	15.5
28	Catch-22	Joseph Heller	Satire	13.25
29	The Handmaid's Tale	Margaret Atwood	Dystopian	14.5
30	Neuromancer	William Gibson	Sci-Fi	15
31	Snow Crash	Neal Stephenson	Sci-Fi	16.25
32	Hyperion	Dan Simmons	Sci-Fi	17
33	Ender's Game	Orson Scott Card	Sci-Fi	14.75
34	I Robot	Isaac Asimov	Sci-Fi	13
35	The Martian	Andy Weir	Sci-Fi	15.5
36	The Silmarillion	J.R.R. Tolkien	Fantasy	18.99
37	American Gods	Neil Gaiman	Fantasy	16.5
38	Neverwhere	Neil Gaiman	Fantasy	13.5
39	Good Omens	Terry Pratchett	Fantasy	14.25
40	The Name of the ...	Patrick Rothfuss	Fantasy	17.25
41	The Fellowship of t...	J.R.R. Tolkien	Fantasy	14.99
42	The Two Towers	J.R.R. Tolkien	Fantasy	14.99
43	The Return of the ...	J.R.R. Tolkien	Fantasy	14.99
44	Lord of the Flies	William Golding	Classic	10.25
45	The Sun Also Rises	Ernest Hemingway	Classic	12.25
46	A Farewell to Arms	Ernest Hemingway	Classic	12.75
47	The Old Man and t...	Ernest Hemingway	Classic	9.75
48	The Grapes of Wrath	John Steinbeck	Historical	14.25
49	Of Mice and Men	John Steinbeck	Classic	10.5
50	East of Eden	John Steinbeck	Historical	16.75

2.

```
/*
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CSC 621
11/5/26
Assignment 5, extract parameters from URL query strings and use those
values to execute a basic database query against a MySQL table, returning
the results directly as JSON or plain text.
*/
const express = require('express');
const pool = require('./mysql');

const app = express();
const PORT = 3300;

function escapeHtml(value) {
    return String(value ?? '')
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#39;');
}

app.get('/', async (req, res) => {
    try {
        const { category, author } = req.query;

        const format = typeof req.query.format === 'string'
            ? req.query.format.toLowerCase()
            : 'json';

        let conditions = [];
        let values = [];

        if (category) {
            conditions.push('category LIKE ?');
            values.push(`%${category}%`);
        }
```

```
        if (author) {
            conditions.push('author LIKE ?');
            values.push(`%${author}%`);
        }

        if (conditions.length === 0) {
            return res.status(400).json({
                error: 'Please provide an author or category search
parameter.'
            });
        }

        const sql = `
            SELECT bookid, title, author, category, price
            FROM books
            WHERE ${conditions.join(' AND ')}
        `;

        const [rows] = await pool.execute(sql, values);

        if (format === 'text') {
            const tableRows = rows.map(book => `
                <tr>
                    <td>${escapeHtml(book.bookid)}</td>
                    <td>${escapeHtml(book.title)}</td>
                    <td>${escapeHtml(book.author)}</td>
                    <td>${escapeHtml(book.category)}</td>
                    <td>${escapeHtml(book.price)}</td>
                </tr>
            `).join('');

            const html = `
                <!DOCTYPE html>
                <html>
                <head>
                    <title>Book Search Results</title>
                </head>
                <body>
                    <h1>Book Search Results</h1>
```

```
                    <table border="1" cellpadding="8">
                        <tr>
                            <th>Book ID</th>
                            <th>Title</th>
                            <th>Author</th>
                            <th>Category</th>
                            <th>Price</th>
                        </tr>
                        ${tableRows || '<tr><td colspan="5">No matching
books found.</td></tr>'}
                    </table>
                </body>
                </html>
            `;

            return res.type('html').send(html);
        }

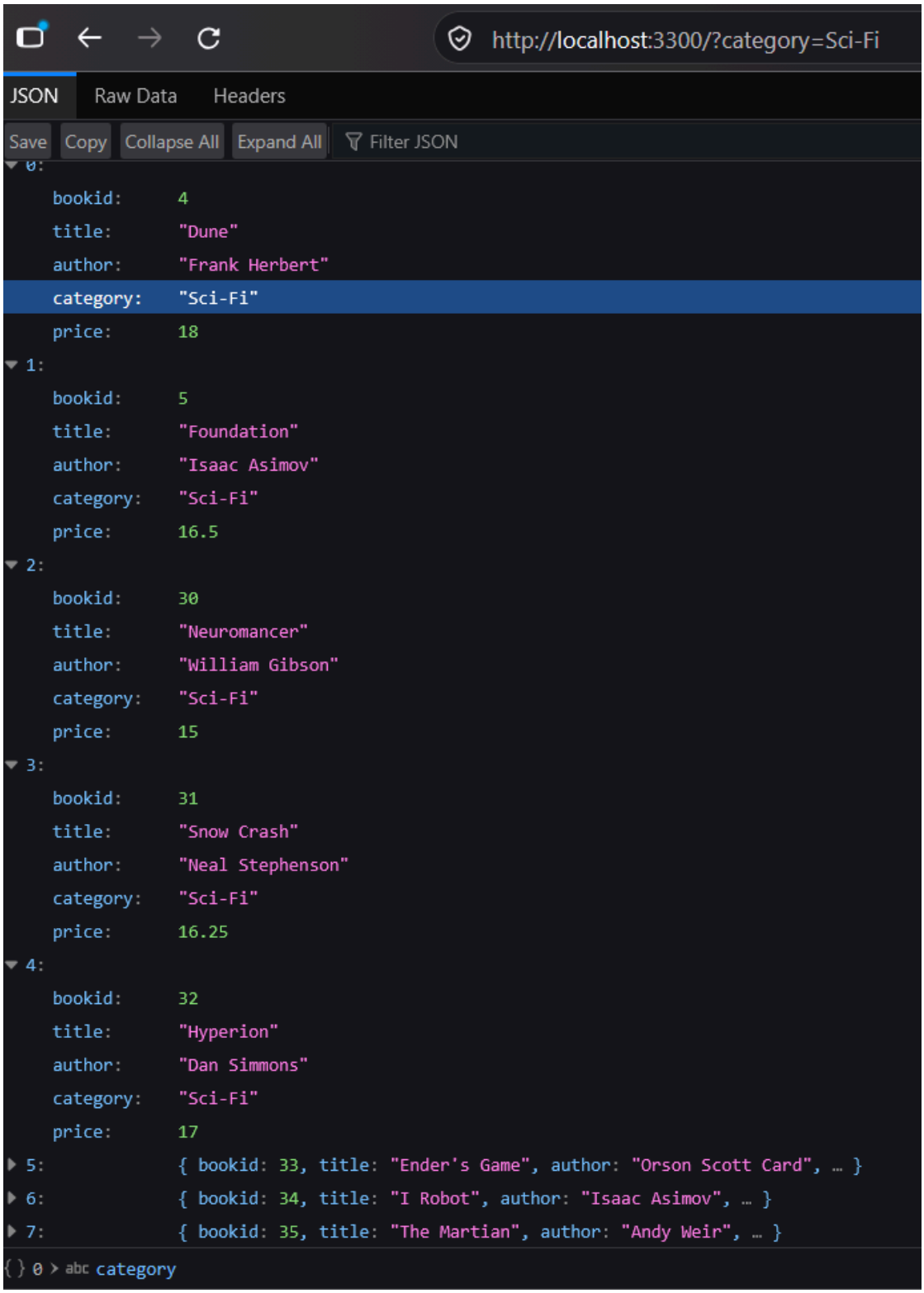
        res.json(rows);

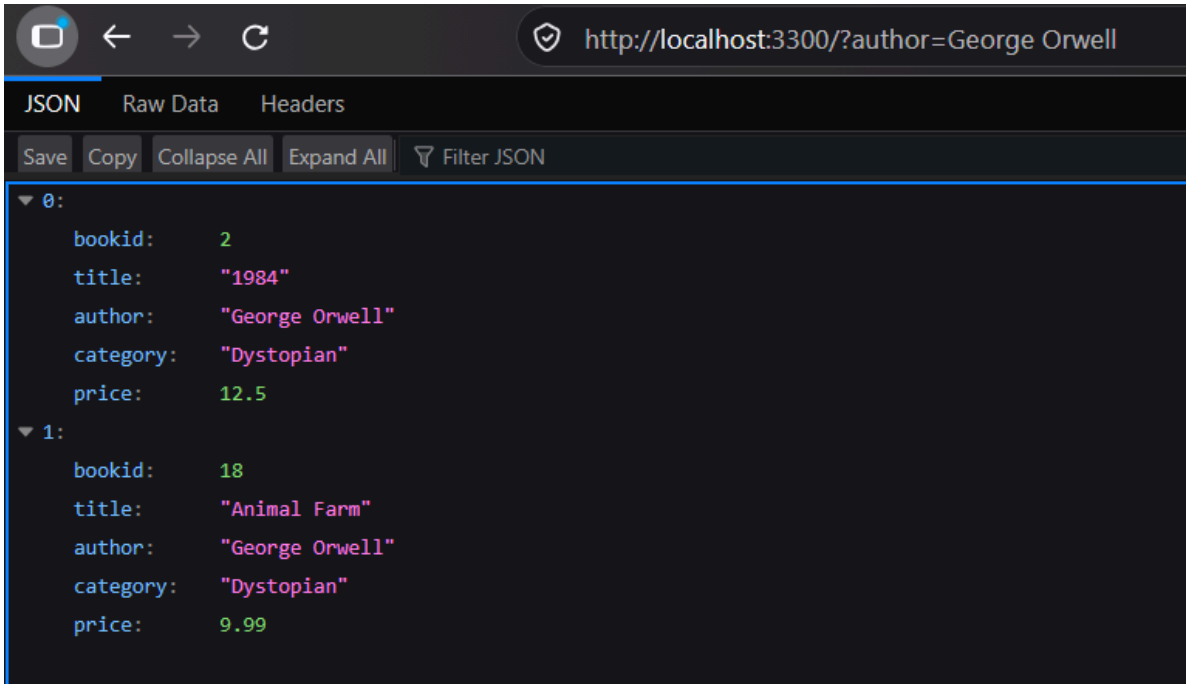
    } catch (error) {
        console.error('Database Error:', error.message);

        res.status(500).json({
            error: 'An error occurred while retrieving books from the
database.'
        });
    }
});

app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
});
```

3.


http://localhost:3300/?category=Sci-Fi
JSON
Raw Data
Headers
Save
Copy
Collapse All
Expand All
Filter JSON
0:
bookid: 4
title: "Dune"
author: "Frank Herbert"
category: "Sci-Fi"
price: 18
1:
bookid: 5
title: "Foundation"
author: "Isaac Asimov"
category: "Sci-Fi"
price: 16.5
2:
bookid: 30
title: "Neuromancer"
author: "William Gibson"
category: "Sci-Fi"
price: 15
3:
bookid: 31
title: "Snow Crash"
author: "Neal Stephenson"
category: "Sci-Fi"
price: 16.25
4:
bookid: 32
title: "Hyperion"
author: "Dan Simmons"
category: "Sci-Fi"
price: 17
5: { bookid: 33, title: "Ender's Game", author: "Orson Scott Card", … }
6: { bookid: 34, title: "I Robot", author: "Isaac Asimov", … }
7: { bookid: 35, title: "The Martian", author: "Andy Weir", … }
{ } 0 > abc category


http://localhost:3300/?author=George Orwell
JSON
Raw Data
Headers
Save
Copy
Collapse All
Expand All
Filter JSON
0:
bookid: 2
title: "1984"
author: "George Orwell"
category: "Dystopian"
price: 12.5
1:
bookid: 18
title: "Animal Farm"
author: "George Orwell"
category: "Dystopian"
price: 9.99